Homo Faber in the making: A cross-cultural analysis of toolmaking skill acquisition in non-industrial societies

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Contents

1 Introduction

Toolmaking, together with bipedalism and language, was once regarded as a defining feature that makes humans distinct from other species (Oakley, 1949). It has also been constantly portrayed as the pinnacle in the evolution of cognition and creativity (e.g., Bergson, 1998; Vaesen, 2012). Although this simple notion of human uniqueness and its associated linear evolutionary narrative was overturned in the past decades by mounting evidence of toolmaking by non-human primates and corvids (Shew, 2017; Shumaker et al., 2024), particularly in the domain of food foraging, the higher-order cases of tool manufacture (i.e., using existing tools to make other tools) as well as the recruitment of toolmaking process for novel functions (Davidson & McGrew, 2005) has not been recorded in any species other than the human lineage. In terms of time allocation of tool-use and toolmaking as well as the level of dependency in caloric intake (Shipman, 2010), toolmaking occupies a central niche in our daily activities as compared with other toolmaking species. On top of the basic survival need, it is also critical to human experience in the sense that a biocultural feedback loop is formed through toolmaking that shapes our minds and bodies in turn (Birch, 2021; Malafouris, 2012, 2021; Stout, 2021), as demonstrated by an extensive corpus of literature revolving around stone tools due to their longevity and great preservation. For example, a series of neuroimaging studies designed by Stout and colleagues (Stout & Chaminade, 2007; Stout & Hecht, 2023; Stout & Khreisheh, 2015) revealed the shared neuroanatomic foundations and differential cognitive demands of various lithic technologies, highlighting the evolutionary importance of perceptual-motor adaptation in technological evolution. Stone toolmaking and use has also been identified as a major selection force driving the diachronic changes in human hand morphology, including both bone and soft tissue anatomy, based on multiple biomechanical/kinematic studies in the context of experimental archaeology (Dunmore et al., 2023; Karakostis et al., 2021; Kivell et al., 2023; Marzke, 2013)

Yet the skill of toolmaking is not born with us — naive individuals need to acquire a given skill either through observing and learning from others (social learning) or repeated trials and errors (individual learning). Very often the skill attainment, particularly this type of motor skill involving extensive hand-eye coordination, requires a combination of these two learning strategies. Although this behavior is of great significance in human evolutionary studies as argued above, the mass machinery manufacturing process of tools introduced by industrialization weaken the necessity of handicraft practices and their inter-generational transmssion in our contemporary daily life. Consequently, ethnographic narratives depicting the subsistence and economy of non-industrial societies, mostly written during the nineteenth and twentieth centuries, provide us a precious opportunity to investigate this topic, among which hunter-gatherer societies represent the most well-studied and well-published sample.

Embedded within the century-old anthropological research tradition revolving around childhood, the learning dynamics of hunter-gatherers have garnered considerable attention in the past two decades due to the burgeoning literature of cultural evolutionary theories (Hewlett et al., 2024). Garfield et al. (Garfield et al., 2016) conducted the first systematic investigation on hunter-gather social learning, involving 982 ethnographic texts from 23 different communities. In this research, they characterized the commonalities of learning behaviors across diverse domains regarding their transmission mode, forms of teaching, and learners' age. Within the acqusition of subsistence skills, which dominated the early and middle childhood, they concluded that oblique transmission becomes more important when compared with vertical transmission with the increasing age of learners. Subsequently, Lew-Levy and her colleagues narrowed down the learner' identities to children and thoroughly examined their learning patterns of subsistence skill, of which toolmaking skill is a subset, depicting a more detailed picture regarding the dominant learning processes and skill types across each developmental periods (Lew-Levy et al., 2017; Lew-Levy, Milks, et al., 2020). Drawing upon both the cross-cultural analysis of existing ethnographies and new systematic field data collected from certain foraging communities, this children-centered research line of inquiry remained to be the most fruitful one in the past few years, emphasizing the evolutionary significance of object/toy as a medium of learning (Lew-Levy et al., 2022; Riede et al., 2023) and the often ignored strategy of peer-learning (Lew-Levy, Kissler, et al., 2020; Lew-Levy et al., 2023). In addition, other researchers conducted systematic observation on the knowledge transmission patterns of societies featuring other forms of subsistence strategies such as the Tsimane forager-farmers (Schniter et al., 2023). More specifically, Schniter et al. (2023) identified the importance of older same-sex relatives and multigenerational pedagogy in the successful reproduction of crucial skills like handicrafts.

Some pressing issues still remain unsolved despite previous intellectual efforts on the topic toolmaking skill acquisition. First and foremost, the extremely low contribution of individual learning in subsistence skills, particularly toolmaking, which according to Garfied et al. (2016) only occupy 3.7% of all cases. There could be several reasons behind it. An obvious one is that ethnographers may systematically care more about communal activities and it's challenging to follow and observe individuals. The other source of bias could be coding. And that what I will try to improve in this project. I have a coding scheme that provide a more detailed evaluation of the learning processes. Second, the unbalanced representation of hunter-gather societies versus all other subsistence types within non-industrial societies mask our understanding because toolmaking (by hand) is not unique to hunter-gatherers.

To understand the diversity and regularity in the process of human toolmaking skill acquisition, this project analyzed 170 non-industrial societies displaying varying subsistence strategies around the world using the eHRAF World Cultures database, with a particular focus on developmental contexts and transmission biases involved in the learning processes of toolmaking skills. At the most basic level, this study provides the most in-depth descriptive investigation on this topic so far (Gerring, 2012). It can help improving the external validity of archaeological experimental design. It can serve as a middle-range framework to better understand the formation of archaeological records. This study investigates the processes of tool-making transmission in non-industrial societies, focusing on fundamental aspects of social learning: who, where, when, how, what, and why. For the "who" aspect, I examined the identities of both the learner and the model, as well as their social relationships, which corresponds to the transmission mode question in cultural evolution. The "where" question addresses the learning context, such as whether it occurs indoors, outdoors within the village, or in an outdoor field. The "when" explores the ages of both the learner and the model. For the "how," I investigated the degree of social interaction involved in the learning process. The "what" focuses on the content of transmission, such as pottery-making, woodworking, or other craft activities. Lastly, the "why" examines the motivations behind a learner's choice of model—whether it is based on the model's recognized expertise, convenience, or familial ties. Gender divisions and their influence on these dynamics are also considered under the "why" question.

Additionally, this study explores whether hunter-gatherer societies differ from other subsistence systems in these traits. By analyzing these variables, I aim to provide a systematic understanding of tool-making transmission and its variability across cultural contexts. Therefore, this paper intends to answer the following three research questions. 1) Do hunter-gatherers have a unique learning pattern compared with groups practicing other types of subsistence strategies? 2) Do learning strategies change across different developmental periods? (More specific hypothesis: will the frequency of social learning decrease through time?) 3) What are some major transmission biases involved? (kin-based? context(success)-based?)

Engaging with the works of Lancy (Lancy, 2017)

2 Methods and materials

0. Scope of study: Global sample (**Figure** 1)

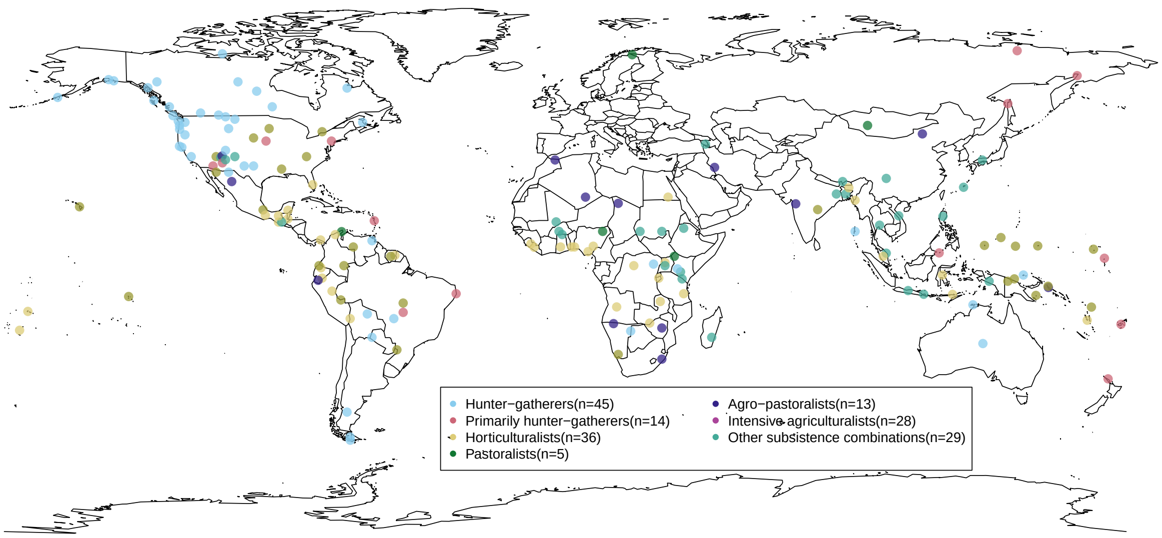

Figure 1: The spatial distribution of non-industrial societies covered in this study (n=169). It should be noted that these coordinate data represent language 'centroids' from the open-access language documentation site glottolog.org instead of the real geographical locations of given communities. These language centroids essentially represent the center point of the area where an relevant language is spoken.

1. EHRAF World Cultures: subject "transmission of skills" + key word "tool". This combination yields ~~133 paragraphs in 96 documents across 72 cultures~~. Another keywords include ~~make (983 paragraphs in 463 documents across 208 cultures)/made (757 paragraphs in 417 documents across 203 cultures)/making (476 paragraphs in 296 documents across 169 cultures)~~, ~~produce (77 paragraphs in 65 documents across 55 cultures)/producing (30 paragraphs in 25 documents across 24 cultures)~~, ~~manufacture (59 paragraphs in 48 documents across 43 cultures)/ manufacturing (5 paragraphs in 5 documents across 5 cultures)~~, ~~create (39 paragraphs in 33 documents across 30 cultures)/ creating (8 paragraphs in 8 documents across 8 cultures)~~, ~~construct (16 paragraphs in 14 documents across 12 cultures)/ constructing (21 paragraphs in 20 documents across 19 cultures)~~, ~~assemble (11 paragraphs~~

~~in 11 documents across 11 cultures)/ assembling (2 paragraphs in 2 documents across 2 cultures)~~, ~~fabricate (3 paragraphs in 3 documents across 3 cultures)/ fabricating (1 paragraphs in 1 documents across 1 cultures)~~, ~~shape (71 paragraphs in 58 documents across 50 cultures)/ shaping (18 paragraphs in 17 documents across 17 cultures)~~, ~~forge (6 paragraphs in 11 documents across 9 cultures)/ forging (6 paragraphs in 6 documents across 5 cultures)~~, ~~form (324 paragraphs in 198 documents across 133 cultures)/ forming (15 paragraphs in 15 documents across 14 cultures)~~.

2. Exclusion criteria: cases of tool use, commercial economy, instance where local people taught the ethnographer how to make tools, instances where it was clearly stated a new form of government-initiated school to save a certain craft, instances where individuals are taught how to make Christianity-related artifacts such altar, and instances where indigenous people mentioned that they learn a specific craft from museum pieces, photographs. However, they may appear as contextual information in the appendix to provide more details of the technologies themselves. archaeological and historical literature are also excluded.

Variables

Important Note: "NA" is used in all variables when there is not enough information to make the relevant inference, which is different from absence. Using the eHRAF database, I employed a dual-system recording approach with multiple variables. Specifically, I first documented the descriptions provided by ethnographers and then categorized the information in a separate column for further analysis.

1. **Area**: using eHRAF's classification system

2. **Society**: using eHRAF's classification system

3. **OWC Code**: The Outline of World Cultures (OWC) is a classification system of the cultures of the world, which is used by eHRAF to organize its databases.

4. **Subsistence**: This variable follows eHRAF's classification system, including Hunter-gatherer, primarily hunter-gatherer, horticulturalist, agro-pastarolist, intensive agriculturalist, other subsistence combinations. The category of commercial economy is excluded.

5. **Year of observation**: The years when the given society was observed by the ethnographers. This information is readily available under the name "field date" in the meta information of eHRAF World Cultures.

6. **Location**: The location where the learning event happens according to the ethnographer's description.

7. **Location (type)**: 1) *indoor:* learning happens inside a building; 2) *outdoor settlement:* learning happens inside the settlement but outside the buildings; 3) *outdoor field*: learning happens outside the settlement

8. **Age of learner**: The learner's age according to the ethnographer's description.

9. **Age group of learner**: 1) *Infancy and Early Childhood* (0-5 years old), 2) *Middle Childhood* (6-12 years old), 3) *Adolescence*(13-16 years old), 4) *Adulthood* (>16 years old). For analytical purpose, this category does not concern the cultural differences on the perception of developmental differences. When a age range is provided, the minimal age will be used in the classification of age group.

10. **Number of learner**: The number of learner according to the ethnographer's description.

11. **Age of model**: The model's age according to the ethnographer's description.

12. **Age group of model**: 1) *Infancy and Early Childhood* (0-5 years old), 2) *Middle Childhood* (6-12 years old), 3) *Adolescence*(13-16 years old), 4) *Adulthood* (>16 years old).

13. **Number of model**: The number of model according to the ethnographer's description.

14. **Transmission modes**: 1) *horizontal transmission*: social learning from individuals of the same generation, age group, or cohort, roughly within 4–5 years of age; 2) *vertical transmission*: children learning from their parents; 3) *oblique transmission*: social learning between individuals of distinct generations or age groups typically from an older generation to a younger generation. It can happen within or beyond extended kin group, such as learning from grandparents, uncles, or other adults in the local population; 4) *mixed*: different transmission modes occurred together in the learning process of a given technology.

15. **Social relationship**: The relationship between the learner and the model according to the ethnographer's description. For example, mother-son/father-daughter/etc.

16. **Transmission bias**: This category follows the framework developed by Kendal at al. (2018). 1) *kin-based bias*: learning a cultural trait from people who are your kin members, 2) *success-based bias*: learning a cultural trait from people because they are successful in this technology, 3) *prestige-based bias*: learning a cultural trait from people because of their high social status; 4) *conformist bias*: learning a specific type of knowledge from other people because it is displayed by the majority of local population; 5) *mixed*: different transmission biases occurred together in the learning process of a given technology.

17. **Learning processes**: 1) *individual learning*: Individual exhibits repeated attempts to learn a skill or develop new skills or knowledge on his or her own, including trial and error and individual practice; 2) *observational learning*: learner directly observes the actions of models and attempts to replicate the observed actions, and no direct interaction between the model and learner is observed or mentioned; 3) *interactive learning*: this category features the verbal, gestural and bodily interactions between models and learners, be it formal ("teaching") or informal ("group play"). It should be noted that these three categories are not mutually exclusive. On the contrary, these three categories are inclusive of each other , where interactive learning is built on both observation and individual practice, and observational learning involves individual practice.

18. **Time spent**: The time spent on learning according to the ethnographer's description.

19. **Time spent (type)**: 1) *one-shot learning*: the learners only need to learn once to acquire a given technology. 2) *repeated learning*: the learners need to multiple time, including individual trial-and-error learning and practice, to acquire a given technology.

20. **Technology**: The subject of learning according to the ethnographer's description.

21. **Raw material**: stone, clay, plant, bones, etc.

22. **Restriction of status involved**: The restriction on the social status involved in the learning according to the ethnographer's description. For example, gender, wealth, religious status.

23. **Note** (habit, context, difficulty perceived by local people and ethnographers, access to raw material)

24. **Reference**

3 Results

4 Qualitative insights

4.1 Protracted and staged learning trajectories

This study partly confirms Dallos's (2021) sharp observation on the significant role of protracted learning in the formation of human culture, as several HRAF ethnographic sources suggested that the learning of toolmaking skills spans multiple developmental periods in an almost structural manner, where each periods feature different approaches to learning and varied task priorities. This applies to scenarios where people learn different different domains of technology at different ages as well as people acquiring one toolmaking skill learn various technological components in a developmentally scaffolding manner. For instance, in Lancy's (1996: 161-162) classical ethnography on Kpelle children development, he noted that:

> "Children learn three types of tasks at three different ages. They begin performing such rudimentary tasks of rice farming as cooking, pounding rice, and chasing away birds at a very early age, usually before they are 9. More difficult tasks, such as hoeing on the farm, net weaving, and trap making, are begun around age 9 and mastered by age 13. Rarer, more complex skills such as cloth weaving, bag weaving, blacksmithing, and medicine making are acquired in the late teens and may be learned after marriage."

Canoe making, a skill that demands not only the acquiring of technical know-how but also the development of proper physical strength to handle heavy logs, best illustrates the case of long training period required for a single toolmaking skill. For example, Wilbert (1976: 360) commented that among the Warao hunter-gatherers' skill training for men last until 25 years old and he further(Wilbert, 1977: 23) wrote that:

> "A boy is associated with canoe-making from early childhood on, initially as spectator, when he accompanies his father and the other men of the family to the forest and watches the production process. Later, as an adolescent, he picks up the axe himself and participates in the process according to his increasing dexterity. By the time he

> gets married, a young man commands the basic skills and may even have a small canoe to call his own."

This pattern of prolonged and staged learning has also recurrently occurred in other ethnographic observations on other indigenous communities such as Chuuk/Truk (Goodenough, 1951: 53-55) in Micronesia, as well as Kapauku (Pospisil, 1958: 40-41) and Trobriands(Austen, 1945: 194-195; Scoditti, 1989: 56) in Melanesia (e.g., "15-20 years of apprenticeship"). However, given the inherent constraint that most ethnographic sources used here only provided a fragmented narrative on craft learning, it is still unclear how common is this mode of protracted learning in the manufacturing of smaller-sized tools that does not necessarily require high level of physical strength as debated in several empirical studies regarding humans' unique life history pattern (Bliege Bird & Bird, 2002; Jones & Marlowe, 2002).

Initiation ceremony is a period for intensive toolmaking skill training, along with other skills required for adulthoold: Laguru, Mende, Eyak, Kaska, Pomo, Maleluka, Tukano, Goajiro, Ona. For example, Hamdani (2001) described the female initiation rites in Luguru, East Africa.

> "The 'mwali' initiation rite...She explained that she literally lived in this hut for a period of three years... It was her mother who trained her to sort out green vegetables, to pound the maize and to prepare food. It was her aunts who taught her how to make sleeping mats, baskets, pottery and even caps. Mama Fulora explained that during her grandmother's times, girls used to learn how to make clothing from barks of trees."

Similarly, Hugh-Jones (1979) provides a detailed narrative regarding the importance of initiation ceremony in toolmaking skill acquisiiton with the example Tukano people.

> "For the initiates, the marginal period is a time of education...Then the elders systematically teach the initiates to make all the different kinds of basketry (carrying baskets, sieves, flat trays, cassava squeezers, etc.)... In addition to basketry, the initiates are taught how to make the feather head-dresses and other ritual ornaments used in dances...Just before the blowing of pepper, the initiates are taught to weave conical fish traps which they go and set in streams in the forest, accompanied by their masori."

4.2 The distribution of expertise

4.3 Causal knowledge in the craftsmanship

Kaska Honigmann & Bennett, 1949

> "Education. Most education for living is received by the child in the course of performing routine activities. Such education is unformalized and undirected... Sometimes generalized instructions are furnished to children who, in carrying these out, often become confused. Thus we observed one fourteen year old girl make tea by putting the leaves into cold water. Nobody referred to the muddy quality of the beverage. The same kind of teaching was used with the ethnographer. Activities were illustrated but never explained in detail, the exposition of principles being ignored."

4.4 Co-production as a pedagogy

Kpelle, Lancy 1984

> "Learning agricultural tasks and pottery-making in the household is part of adjusting to membership in the community. Parents are responsible for teaching the value of industriousness and are held accountable throughout the life of the child. Agricultural tasks are scheduled in a yearly cycle that provides sufficient flexibility for a man to carry out community and familial ceremonial responsibilities."

Garo Burling 1963

> "Only after he is about twelve years old does anything like formal teaching take place. The most difficult skill to learn or to teach is basket-making, which the father or some other man must impart to a boy. The instruction consists mostly of telling a child to make a particular basket as he has watched others do and then reprimanding him if he does it wrong. Occasionally a man will demonstrate, but I never saw anyone who was able to analyze and point out the mistakes another was making. If a person makes a mistake in a complex basket, for instance, his instructor will take the basket away, redo it himself, and then tell him to do it that way henceforth. No effort is

made to break the complex weaving process down into steps so as to make it easy to understand and learn systematically, although it is recognized that certain baskets are easier to weave than others. Fathers tell their sons to make the easier types first, and then gradually lead them to the more difficult ones."

Imamura and Akiyama (2016): San foragers older children teach tool manufacture skills to younger children through both example and correction.

4.5 Skill training: A byproduct of socialization

Nash 1970 Tzeltal

> "Learning agricultural tasks and pottery-making in the household is part of adjusting to membership in the community. Parents are responsible for teaching the value of industriousness and are held accountable throughout the life of the child. Agricultural tasks are scheduled in a yearly cycle that provides sufficient flexibility for a man to carry out community and familial ceremonial responsibilities."

5 Conclusion

Who: People learn most toolmaking skills from their close kins, particularly parents. Extended family members like grandparents and uncles/aunts are also heavily involved in this process. When For many skills, they started the learning processes in middle childhood (6-12 years old), which can last for several years. The initiation ceremonies provide opportunities for a period of intensive training. How: Although individual practice and trial and error always exist, toolmaking skill acquisition involves lots of social interactions, during which causal knowledge is rarely explicitly transmitted. Where: Learning locations are dependent on the specific task involved. Why: Toolmaking skill transmission is regarded as an important way of community integration. What: The three most frequent toolmaking skills are clothes making, basket making, and pottery making.

I see it as a stepping stone instead of a definitive text for a better understanding of the subject matter, which merely serves as a starting point for an open dialogue and a project to be collaborated with future researchers like the "Play Object Play" database (Riede et al., 2023).

This paper also provides many empirical insigits into some key debates or theories in anthropological discourses, such as the existence of teaching, community of practice, .

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